


# *Topic Detection in Scholarly Documents*

**Angelo Salatino**

KMi, The Open University, UK

angelo.salatino@open.ac.uk

**@angelosalatino.bsky.social**

**SESAME 2026**

Knowledge Media Institute

The Open University

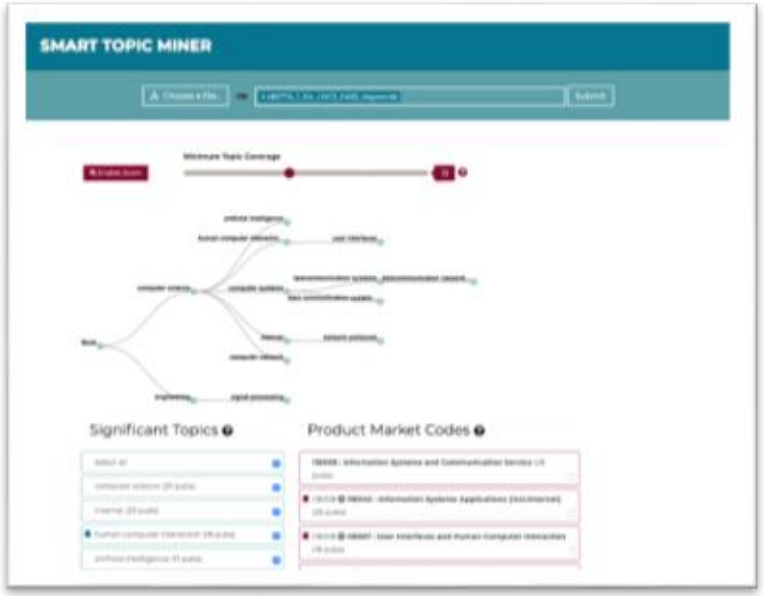
# Promise

- History
- Computer Science Ontology and the CSO Classifier
- Use of CSO Classifier
- KOS of Research Fields
- Topic Classification for UKRI
- Lesson Learnt

# Why topic-annotated documents are important?

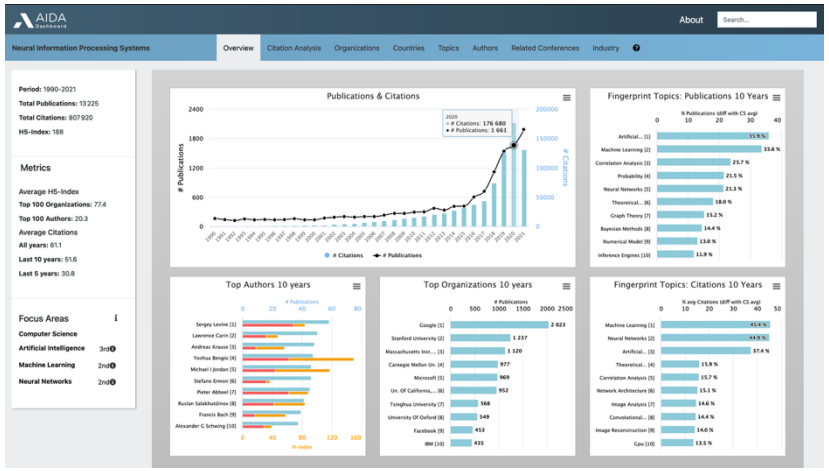
## Digital Libraries

- Organising information
- Augmented retrieval
- Recommender systems
- Integration and interoperability
- Knowledge management and preservation




## Information Science

- Analysis of the research landscape
- Trend analysis and forecasting
- Team composition
- Research Impact



# History in pills

- **Manual and Keyword indexing (~1960)**
  - Early systems relied on human expert, journal categories, and simple keyword matching
- **Statistical Topic Modelling (~1990)**
  - Based on LDA (or similar approaches) which automated topic discovery using document-word probabilities
- **Knowledge Graph and Ontologies (~2010)**
  - Structured networks that map research concepts through semantic relationships
- **Deep Learning and Transformers (~2018)**
  - Dense embeddings and LLMs for context-aware topic extraction

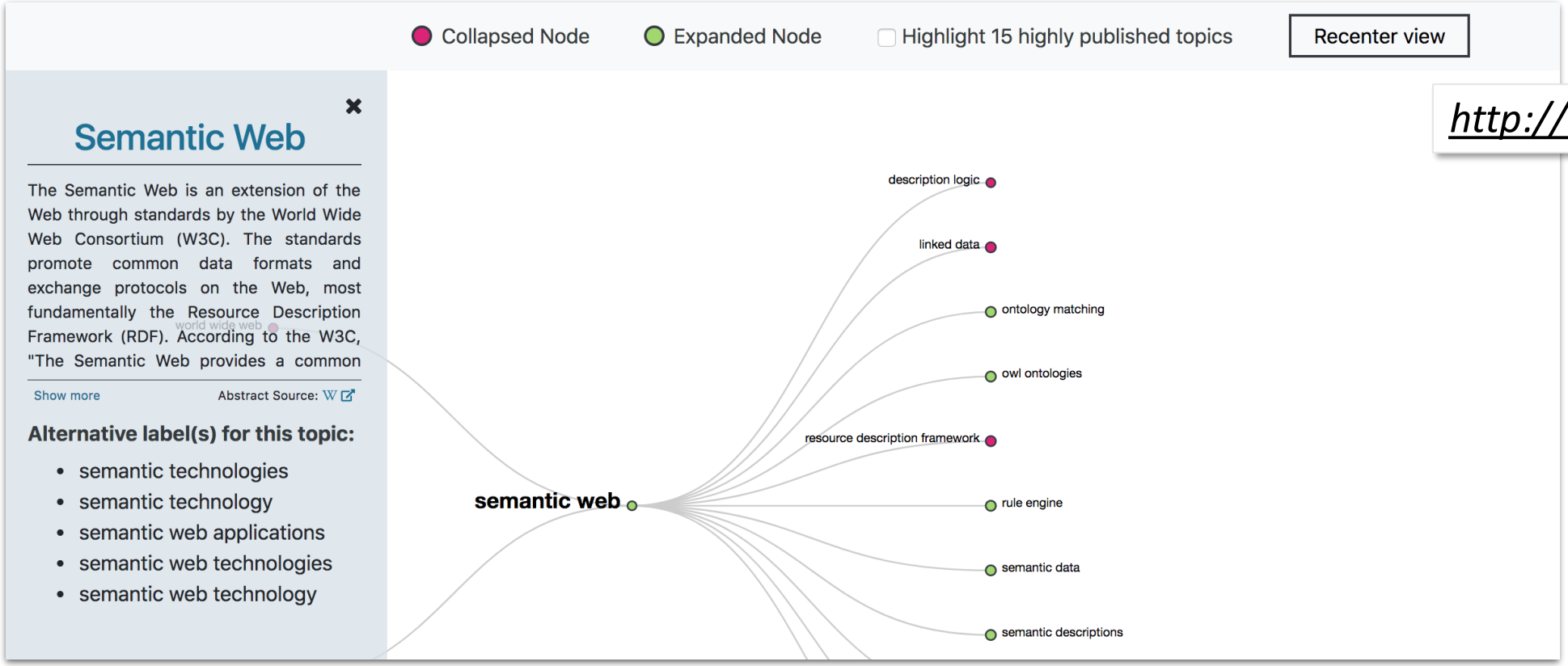


# CSO Classifier

Yutong Liu & Kingston School of Art / Better Images of AI / Talking to AI / CC-BY 4.0

# Our journey: knowing the territories of research

The Computer Science Ontology (CSO) is a large-scale, automatically generated ontology of research areas. It is the largest ontology in the field of Computer Science, including about 15K topics and 166K semantic relationships.

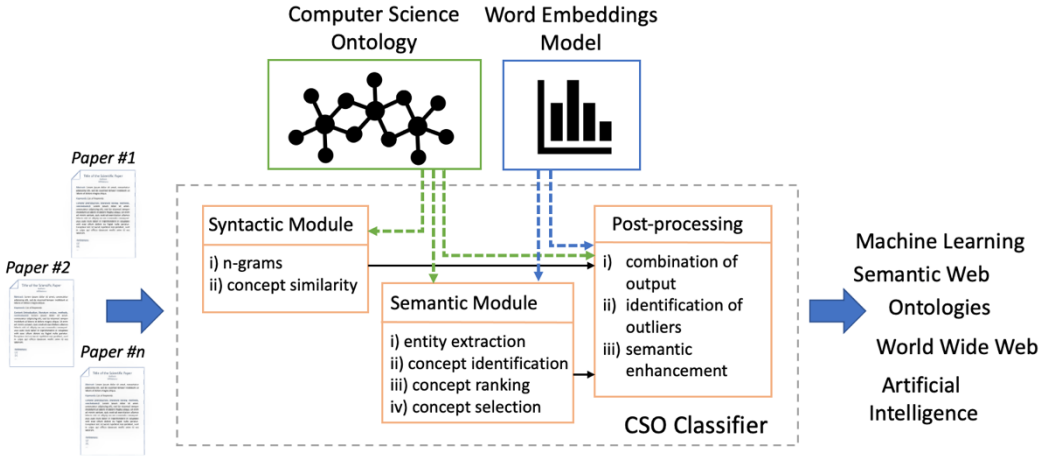



http://w3id.org/cso

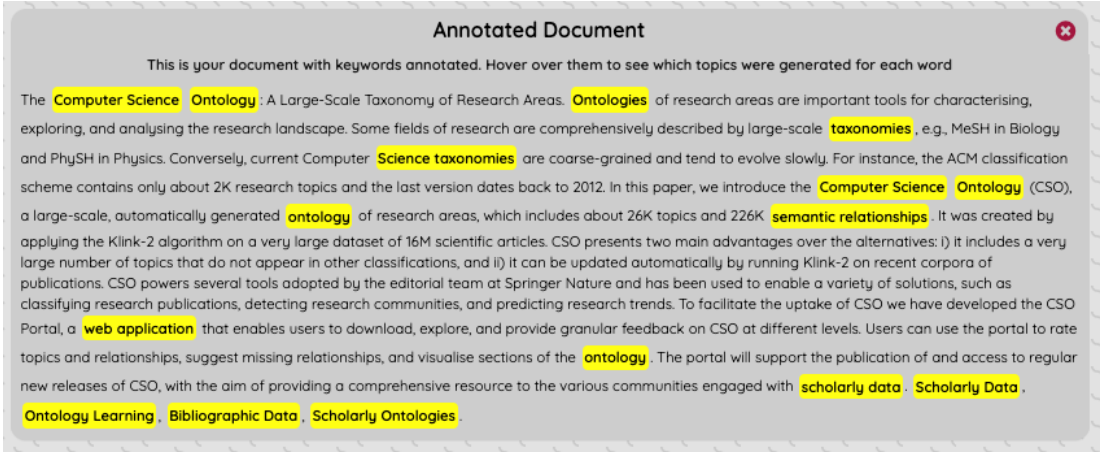
*Salatino et al. (2018) "The computer science ontology: a large-scale taxonomy of research areas."* ISWC 2018

# Classification of Research Outputs - The CSO Classifier

The **CSO Classifier** is an unsupervised approach for automatically classifying documents according to CSO. It is being used by several organisations to classify articles, scientific reports, and courses.




Python Library: https://pypi.org/project/cso-classifier/

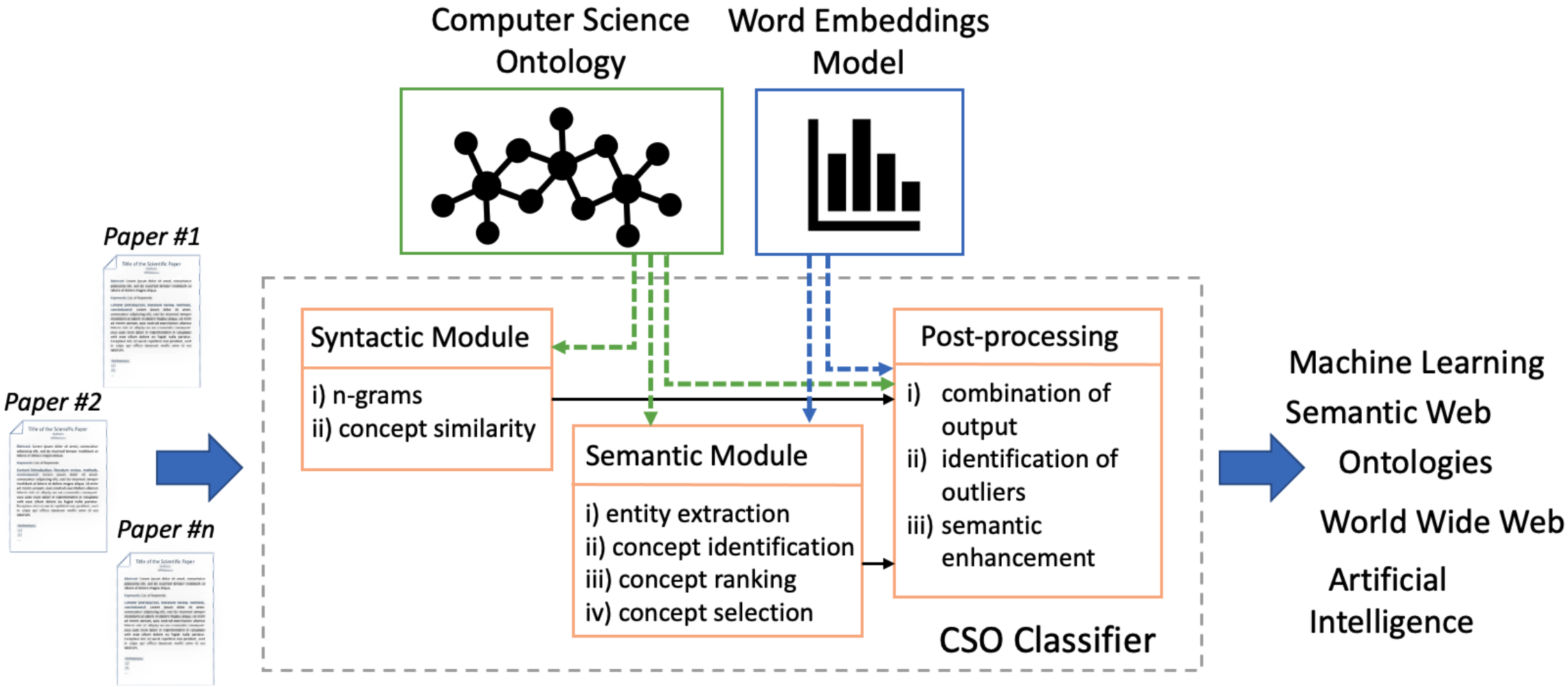



**Version 4.0** of the CSO Classifier recently released!

SPRINGER NATURE

*Salatino et al*. (2019) *"The CSO classifier: Ontology-driven detection of research topics in scholarly articles."* TPDL 2019

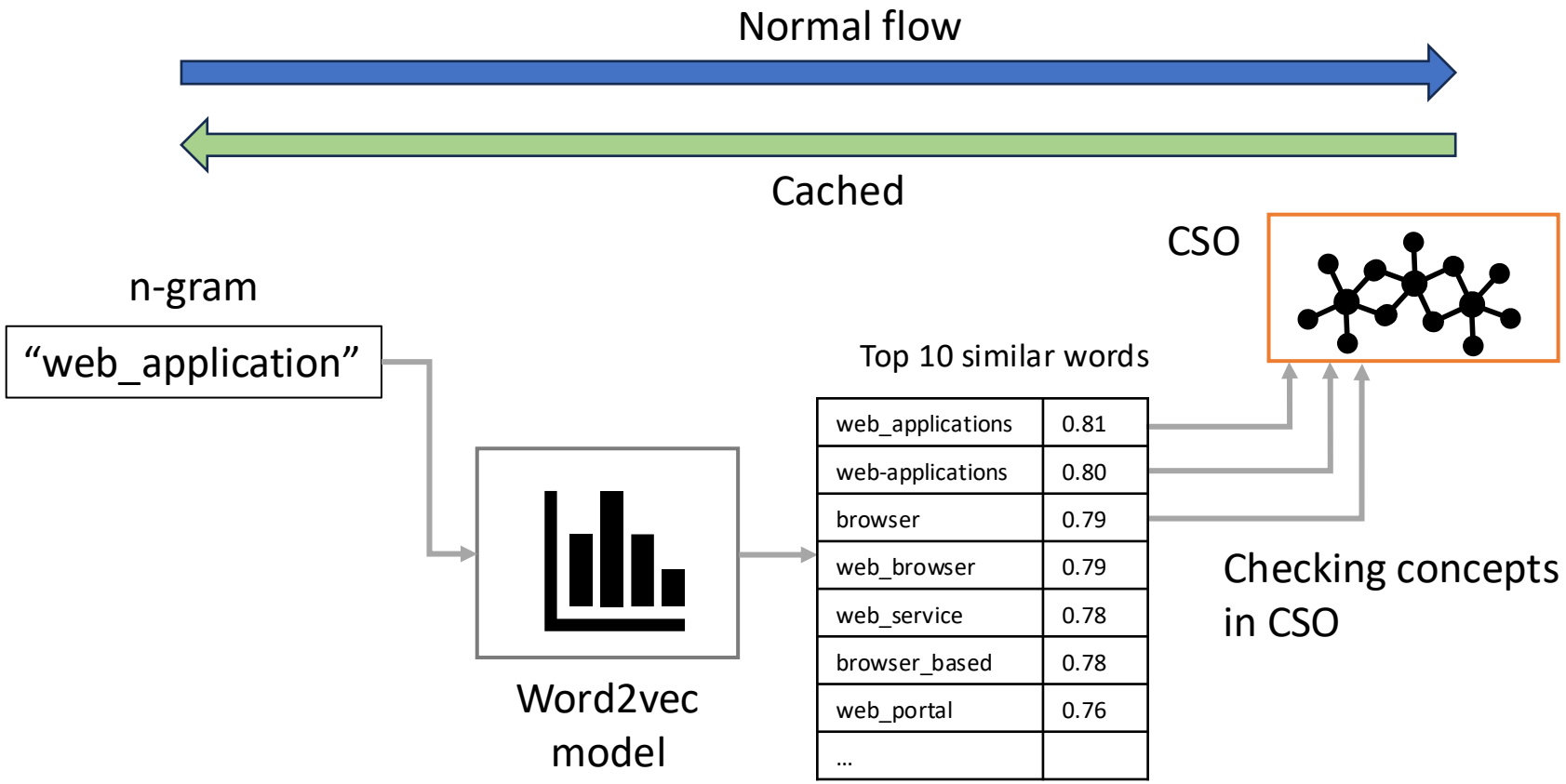
# Let's dissect the classifier



# Need for Speed: caching the model

SPRINGER NATURE

- From topics, we identified all words in the model vocabulary that could trigger them



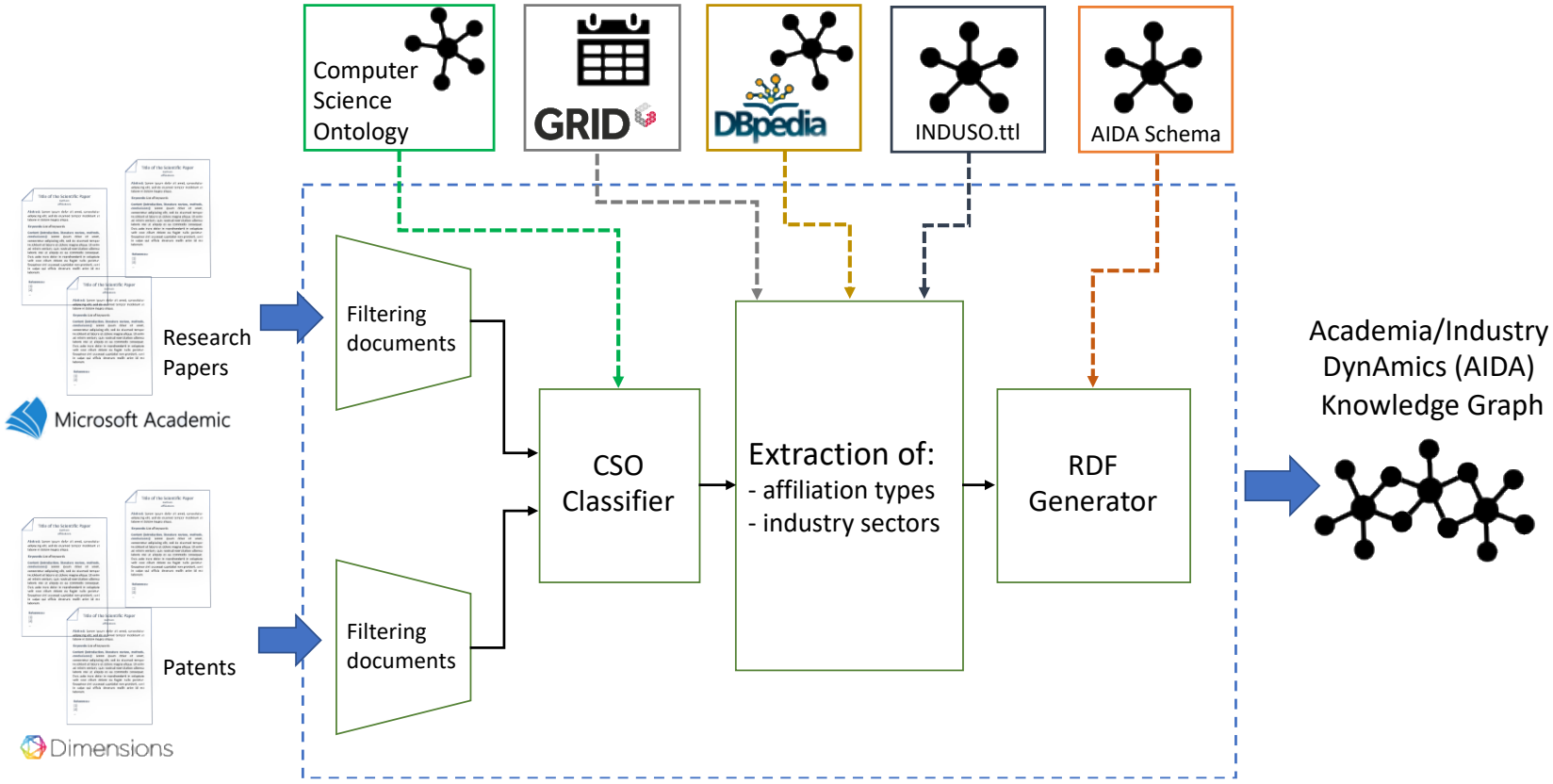


# Using the CSO Classifier in Metascience

Yutong Liu & Kingston School of Art / Better Images of AI / Talking to AI / CC-BY 4.0

# AIDA Knowledge Graph

A large knowledge graph describing **25M publications and 8M patents** classified in terms of research topics, authors' affiliations (academy, industry, or collaborative), and industrial sectors (e.g., automotive).

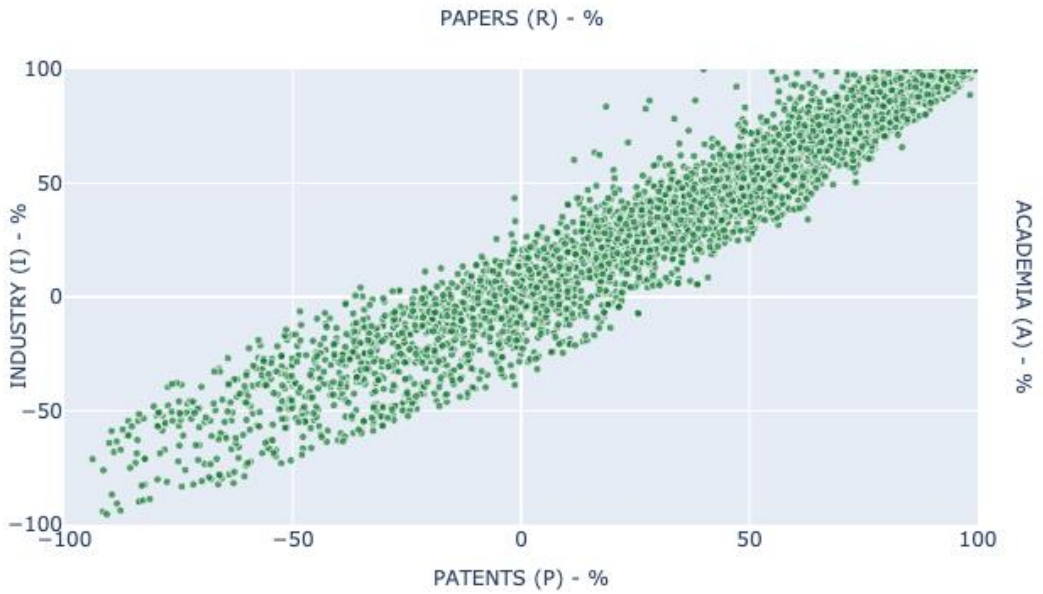


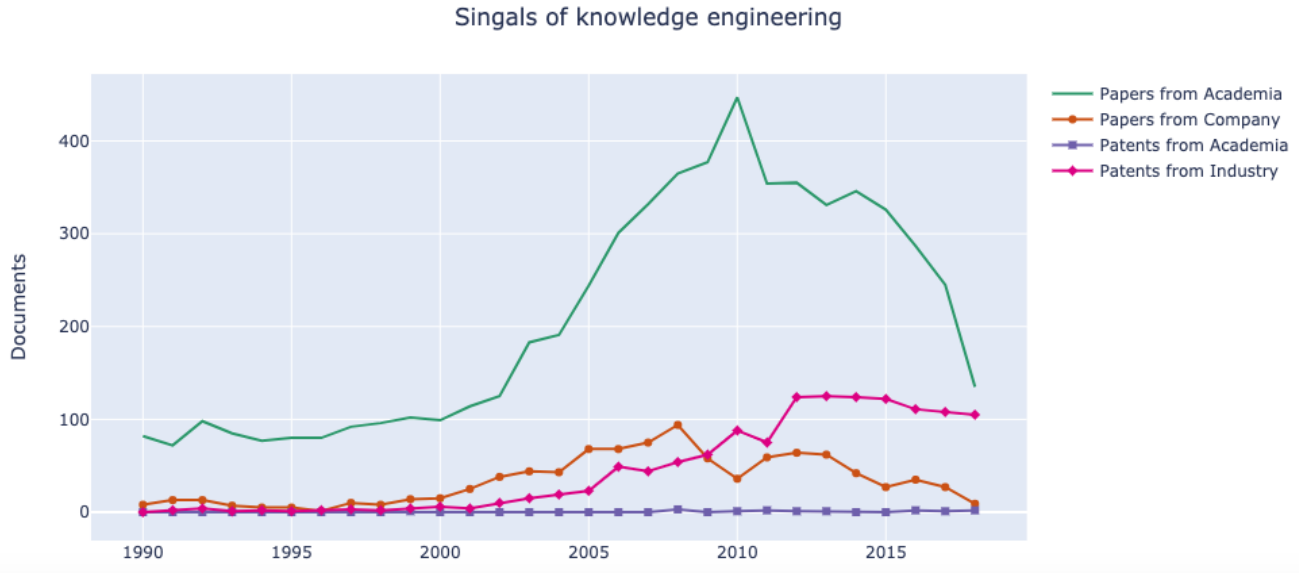

http://w3id.org/aida

*Angioni et al. (2021). AIDA: A knowledge graph about research dynamics in academia and industry. Quantitative Science Studies, 2(4).*

# ResearchFlow

A system for **analysing the flow of knowledge** between academia and industry and **generates analytics**, such as topic evolution, emerging trends, and impact forecasting, by leveraging topic signals extracted from the AIDA KG.



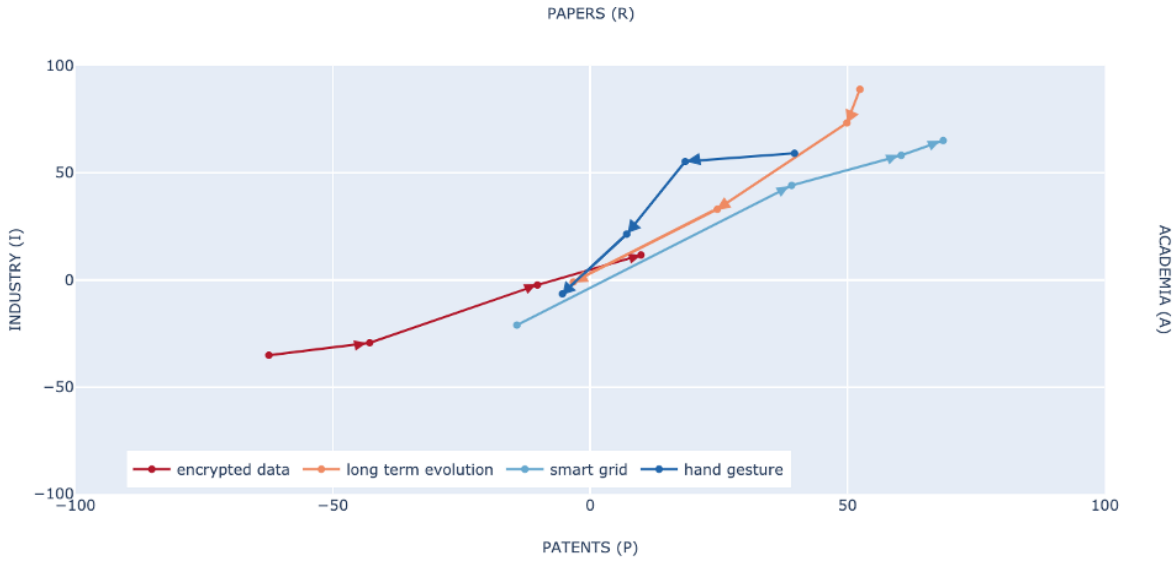

Distribution of topics according to **RP** and **AI** in 2007-18

*Salatino et al. (2020) "Researchflow: Understanding the knowledge flow between academia and industry."* EKAW 2020.

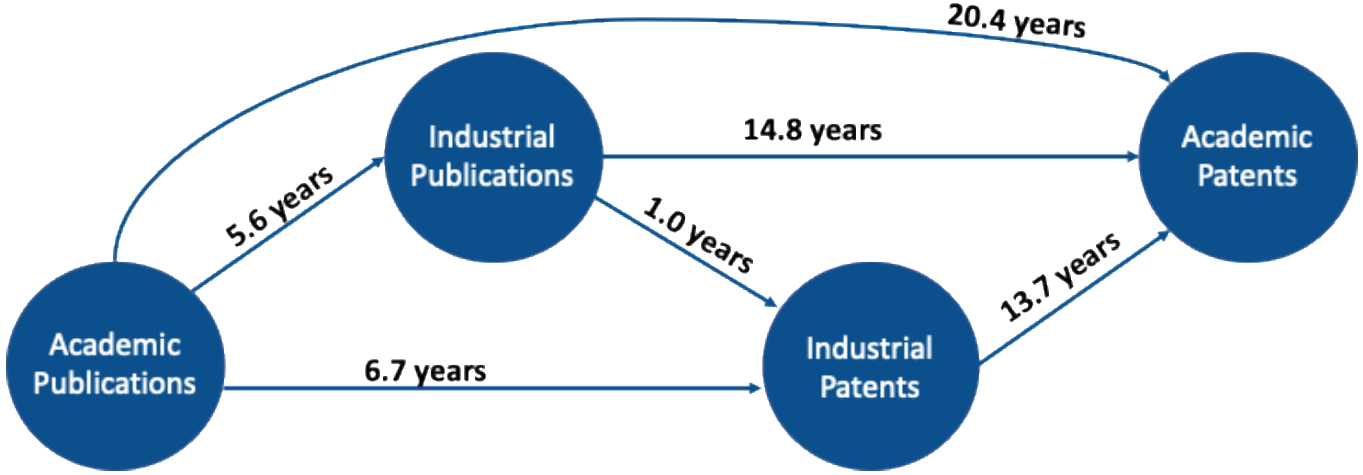
# Obtained insights

We assessed which signal precedes another in addressing a certain topic

- A topic is considered emergent when it is associated with at least 10 documents
- We computed the time lag between signals

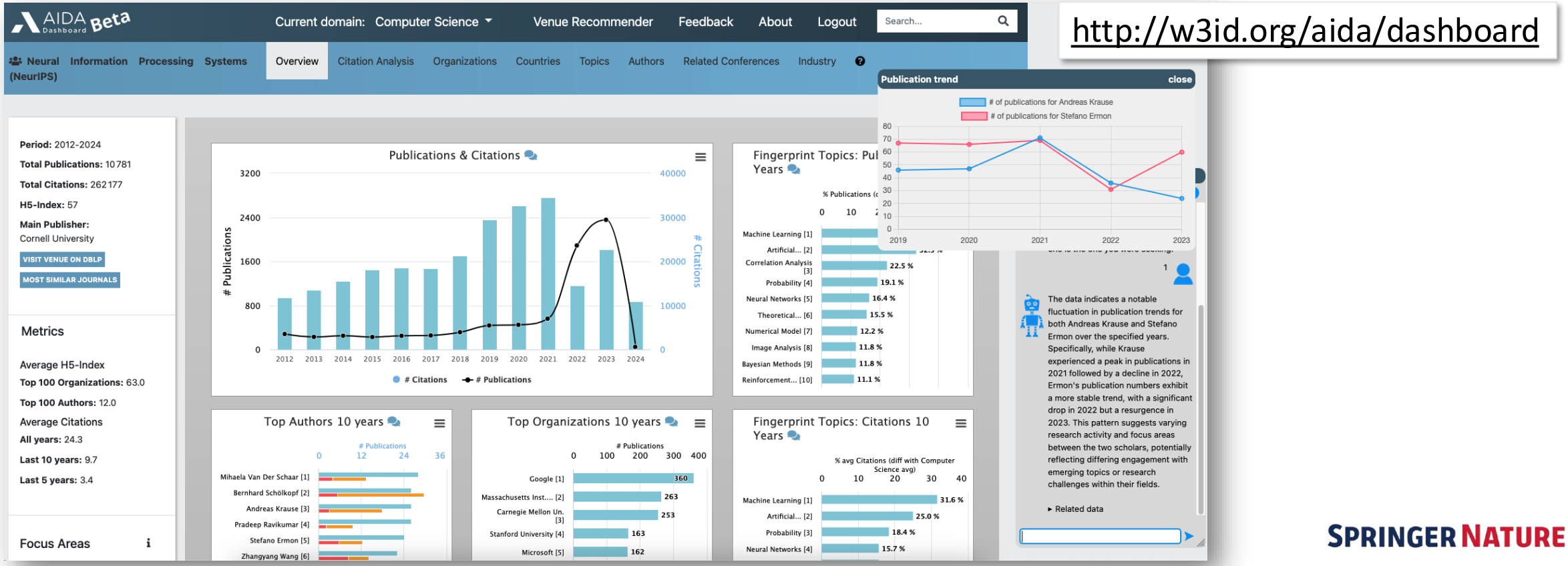



| Topics | Appear first in |
|---|---|
| 89.8% | Academic publications |
| 3.0% | Industrial publications |
| 7.2% | Industrial patents |
| none | Academic patents |



# AIDA Dashboard

The AIDA Dashboard is a tool for exploring and comparing **scientific venues**. It supports advances analyses of **research trends** and relevant **industrial sectors**.




http://w3id.org/aida/dashboard

SPRINGER NATURE

*Angioni et al. (2022) Leveraging Knowledge Graph Technologies to Assess Journals and Conferences at Springer Nature. ISWC 2022.*

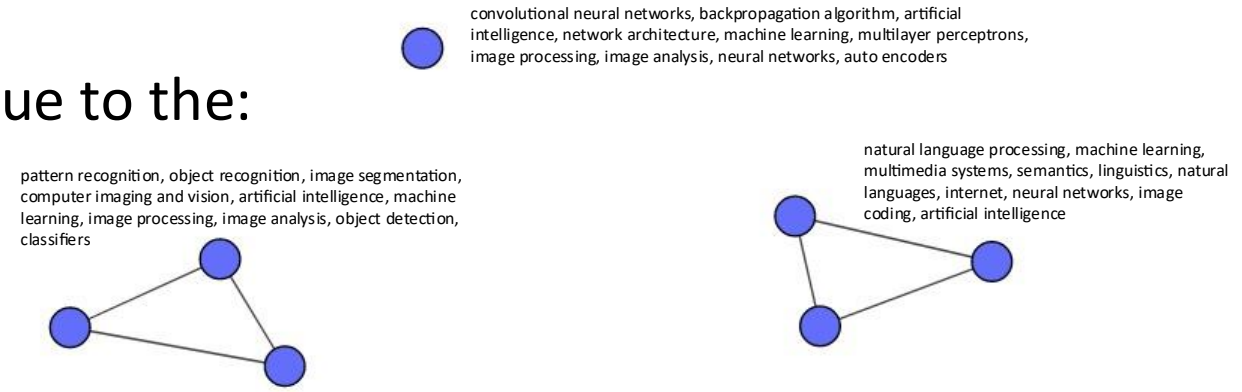
# Diversity of Expertise

Team Diversity has been studied across several dimensions:

- Nationality, Ethnicity, Institution, Gender, Academic age, Disciplinary Background, Team size
- Positive correlation between diversity and impact

No study on diversity of expertise, mainly due to the:

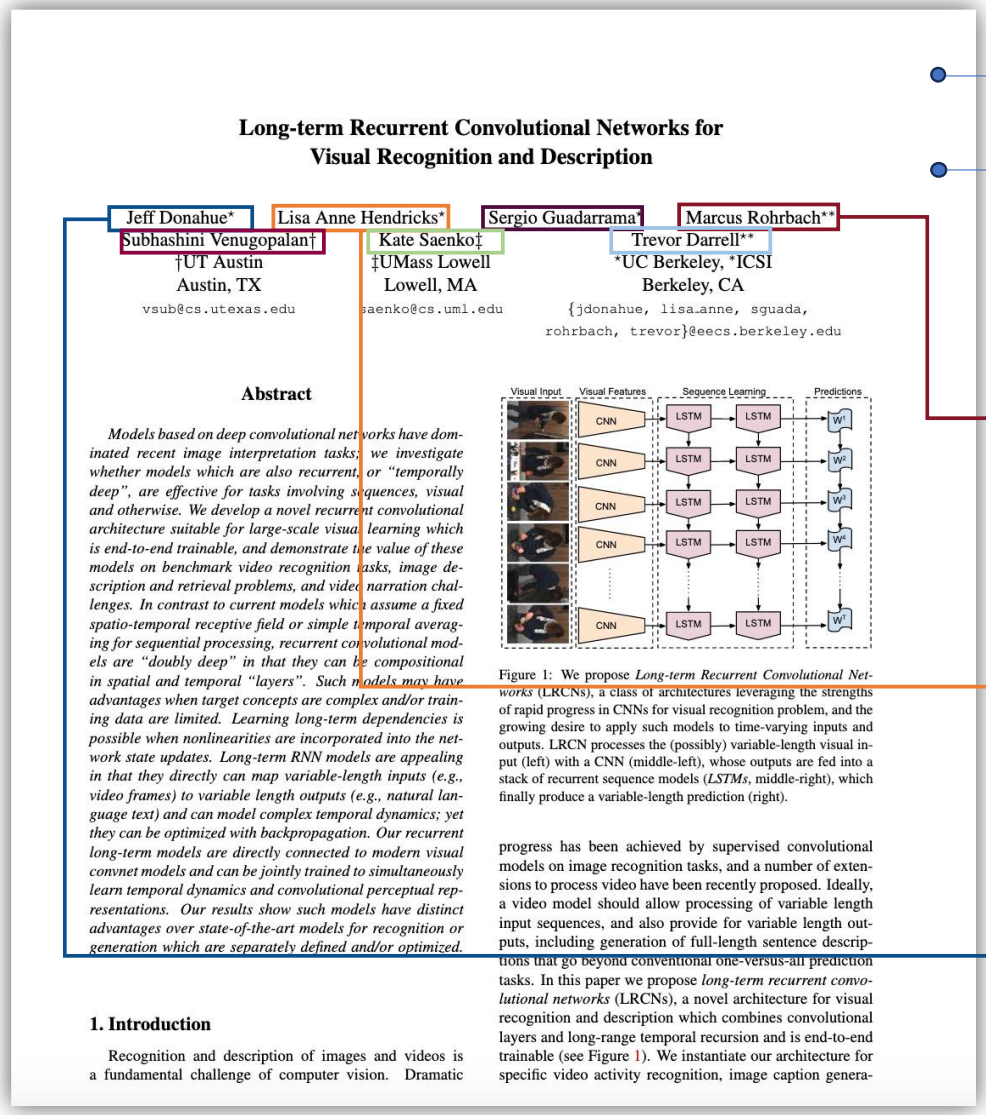
- Lack of controlled vocabulary
- Tool to extract expertise




Can **diversity of expertise** within a research team **influence the number of citations** gathered in the first five years?

*Salatino et al. (2026) "Does Diversity of Expertise Drive Citation Impact? Evidence from Computer Science." Scientometrics: 1-28.*

# The approach: a glimpse



**Long-term Recurrent Convolutional Networks for Visual Recognition and Description**


Jeff Donahue* Lisa Anne Hendricks* Sergio Guadarrama† Marcus Rohrbach**
Subhashini Venugopalan† Kate Saenko‡ Trevor Darrell**
†UT Austin ‡UMass Lowell *UC Berkeley, *ICSI
Austin, TX Lowell, MA Berkeley, CA
vsub@cs.utexas.edu saenko@cs.uml.edu {jdonahue, lisa_anne, sguada, rohrbach, trevor}@eecs.berkeley.edu



**Abstract**

*Models based on deep convolutional networks have dominated recent image interpretation tasks; we investigate whether models which are also recurrent, or "temporally deep", are effective for tasks involving sequences, visual and otherwise. We develop a novel recurrent convolutional architecture suitable for large-scale visual learning which is end-to-end trainable, and demonstrate the value of these models on benchmark video recognition tasks, image description and retrieval problems, and video narration challenges. In contrast to current models which assume a fixed spatio-temporal receptive field or simple temporal averaging for sequential processing, recurrent convolutional models are "doubly deep" in that they can be compositional in spatial and temporal "layers". Such models may have advantages when target concepts are complex and/or training data are limited. Learning long-term dependencies is possible when nonlinearities are incorporated into the network state updates. Long-term RNN models are appealing in that they directly can map variable-length inputs (e.g., video frames) to variable length outputs (e.g., natural language text) and can model complex temporal dynamics; yet they can be optimized with backpropagation. Our recurrent long-term models are directly connected to modern visual convnet models and can be jointly trained to simultaneously learn temporal dynamics and convolutional perceptual representations. Our results show such models have distinct advantages over state-of-the-art models for recognition or generation which are separately defined and/or optimized.*





Figure 1: We propose *Long-term Recurrent Convolutional Networks* (LRCNs), a class of architectures leveraging the strengths of rapid progress in CNNs for visual recognition problem, and the growing desire to apply such models to time-varying inputs and outputs. LRCN processes the (possibly) variable-length visual input (left) with a CNN (middle-left), whose outputs are fed into a stack of recurrent sequence models (*LSTMs*, middle-right), which finally produce a variable-length prediction (right).

**1. Introduction**

Recognition and description of images and videos is a fundamental challenge of computer vision. Dramatic progress has been achieved by supervised convolutional models on image recognition tasks, and a number of extensions to process video have been recently proposed. Ideally, a video model should allow processing of variable length input sequences, and also provide for variable length outputs, including generation of full-length sentence descriptions that go beyond conventional one-versus-all prediction tasks. In this paper we propose *long-term recurrent convolutional networks* (LRCNs), a novel architecture for visual recognition and description which combines convolutional layers and long-range temporal recursion and is end-to-end trainable (see Figure 1). We instantiate our architecture for specific video activity recognition, image caption genera-

#citations 5yrs: 2652

Publication Year: 2015







Papers in the 5 years before collaboration

```
['multimedia systems', 'natural languages', 'image processing', 'machine learning', 'artificial intelligence', 'semantics', 'ontology', 'natural language processing', 'image coding', 'linguistics']
```

```
['convolutional neural networks', 'auto encoders', 'multilayer perceptrons', 'image analysis', 'image processing', 'machine learning', 'neural networks', 'artificial intelligence', 'network architecture', 'backpropagation algorithm']
```

```
['computer imaging and vision', 'object detection', 'image analysis', 'image processing', 'pattern recognition', 'machine learning', 'artificial intelligence', 'neural networks', 'object recognition', 'image segmentation']
```

Author Expertise

# Results: Number of sub-teams

| Group Identifier | Citation ranges (c) | Citation Median | low | moderate | high | very high | (very high + high) / low |
|---|---|---|---|---|---|---|---|
| A | 2 ≤ c < 5 | 3 | 65.18% | 32.00% | 2.60% | 0.23% | 4.34% |
| B | 5 ≤ c < 10 | 6 | 62.77% | 33.77% | 3.14% | 0.32% | 5.52% |
| C | 10 ≤ c < 15 | 12 | 61.17% | 34.68% | 3.72% | 0.44% | 6.79% |
| D | 15 ≤ c < 20 | 17 | 60.07% | 35.44% | 4.02% | 0.48% | 7.48% |
| E | 20 ≤ c < 30 | 24 | 59.01% | 36.12% | 4.37% | 0.50% | 8.26% |
| F | 30 ≤ c < 40 | 34 | 58.09% | 36.57% | 4.80% | 0.54% | 9.19% |
| G | 40 ≤ c < 50 | 44 | 58.06% | 36.20% | 5.11% | 0.62% | 9.87% |
| H | 50 ≤ c < 100 | 64 | 56.29% | 37.22% | 5.69% | 0.80% | 11.53% |
| I | 100 ≤ c < 150 | 118 | 54.92% | 37.63% | 6.37% | 1.08% | 13.57% |
| J | c ≥ 150 | 226 | 53.71% | 37.57% | 7.24% | 1.48% | 16.25% |
| ALL | | | 62.13% | 33.99% | 3.49% | 0.39% | 6.23% |

- % Papers with low diversity is inversely correlated ($r=-0.8$, $p=0.03$)
- % Papers with high diversity is positively correlated ($r=0.97$, $p<0.0001$)
- Ratio between highs (*high+very high*) and low is highly correlated ($r=0.9$, $p<0.0001$)

# Why this matters?

The data suggests that in Computer Science, teams comprising members with varied technical backgrounds are a primary driver of high-impact research

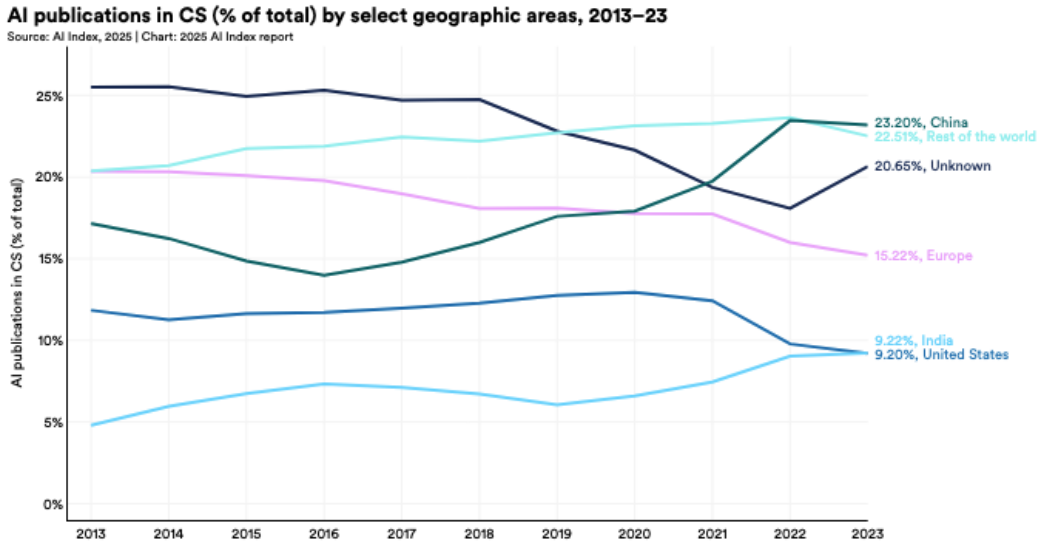
- **For Policy Makers**: These findings suggest that funding and institutional support should **prioritise interdisciplinary and cross-domain** collaborations.
  - Policy should encourage the assembly of teams with complementary, diverse skill sets to tackle complex problems (instead of focusing just on team size).

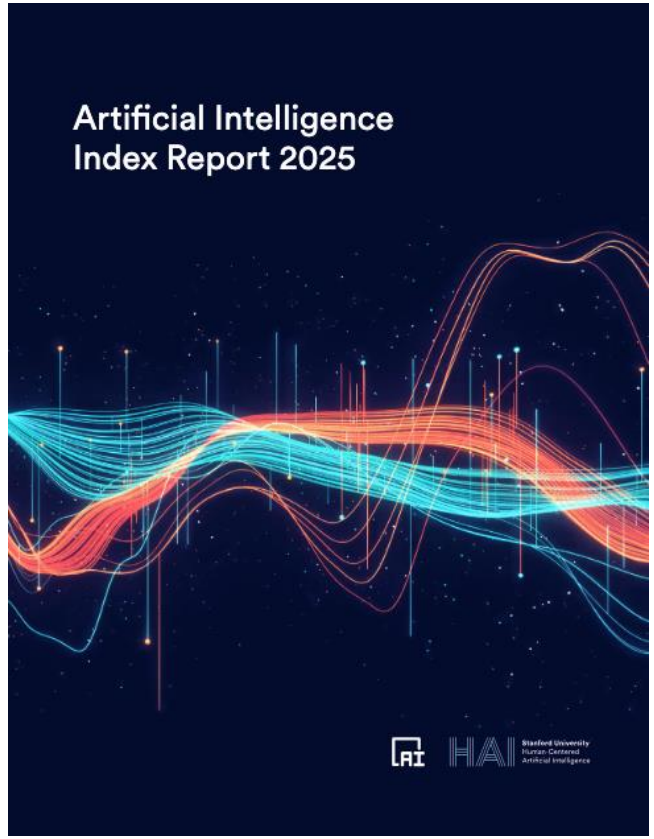
- **For Metascience**: Measuring expertise has been a long-standing challenge. With this framework, we provide a tool for better understanding the mechanics of scientific innovation and how cross-pollination of domains lead to breakthroughs.

# Stanford AI Index Report (2025 & 2026)

A primary Artificial Intelligence reference for global policymakers (UK, US, and EU) and major media outlets like The Guardian, Bloomberg, and The New York Times.

- analyses to benchmark **AI growth** against the broader Computer Science field, using **CSO** and **CSO Classifier**
- **publication and citation trends**, categorised by geographic region, institutional sectors (industry/government), venue, and subdisciplines.




https://hai.stanford.edu/ai-index/2025-ai-index-report

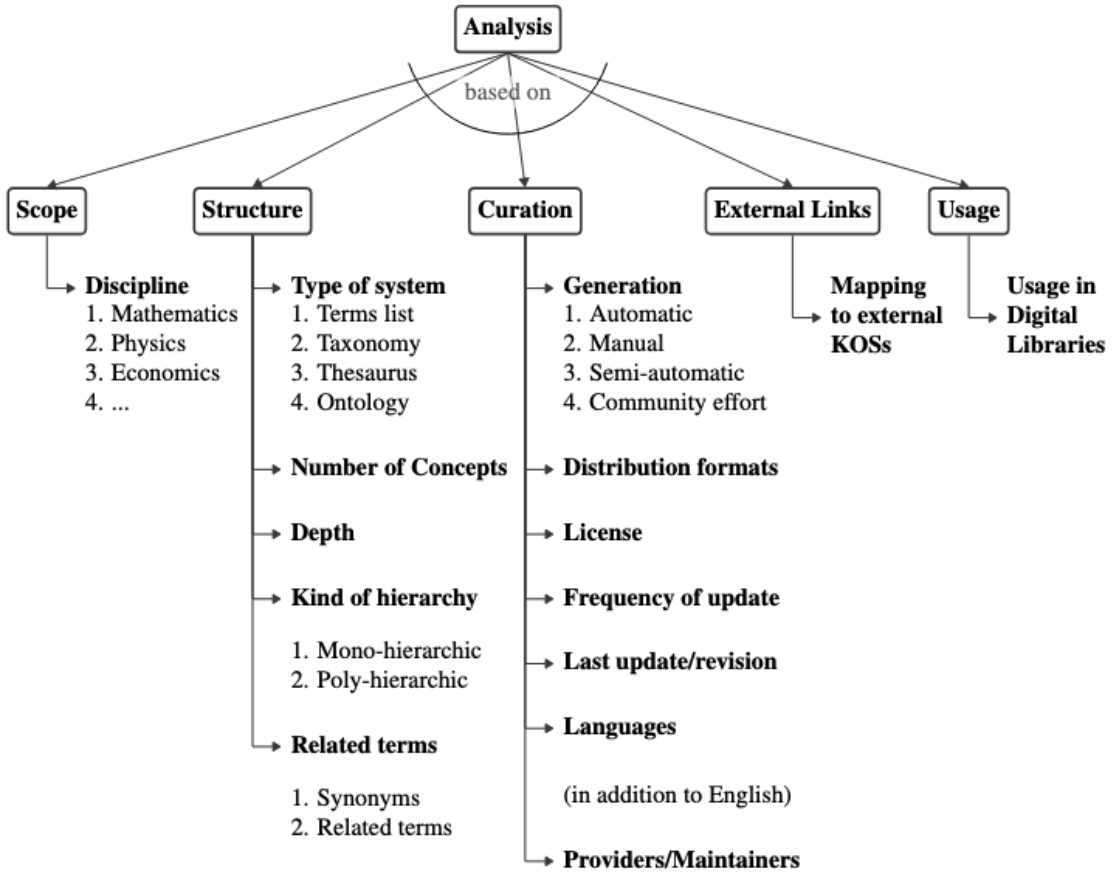
In 2025, we supported the Stanford team in developing all the scientometric analytics.




# Beyond Computer Science?

Rick Payne and team / Better Images of AI / AI is... / CC-BY 4.0

# Knowledge Organization Systems of Research Fields




| Knowledge Organization System | Main Discipline | #Concepts | Depth | Sy | Hr | RT |
|---|---|---|---|---|---|---|
| Agrovoc Thesaurus | Agriculture | 41K | 14 | O | P | Y |
| Art and Architecture Thesaurus | Art & Architecture | 58K | 13 | O | P | Y |
| EDAM | Bioinformatics | 264 | 7 | O | P | Y |
| Open Biological and Biomedical Ontology | Biology | 5.3M | 39 | O | P | Y |
| Biomedical Ontologies from BioPortal | Biomedicine | 13M | - | O | P | Y |
| ChemOnt | Chemistry | 4825 | 11 | O | M | Y |
| ACM Computing Classification Scheme | Computer Science | 2114 | 6 | T | P | N |
| Computer Science Ontology | Computer Science | 14K | 13 | O | P | Y |
| Computer Science Subject Headings from Wikipedia | Computer Science | 7354 | 20 | T | M | N |
| Journal of Economic Literature | Economics | 859 | 3 | T | M | N |
| STW Thesaurus for Economics | Economics | 14K | 13 | O | P | Y |
| IEEE Thesaurus | Engineering | 12K | 3 | U | P | Y |
| GeoRef Thesaurus | Geology | 33K | 5 | U | M | Y |
| U.S. Geological Survey Library Classification System | Geology | 2k | 6 | T | M | N |
| Mathematics Subject Classification | Mathematics | 6598 | 3 | T | M | Y |
| Medical Subject Headings | Medicine | 30K | 13 | O | P | Y |
| National Library of Medicine classification | Medicine | 4761 | 4 | T | M | N |
| Unified Medical Language System | Medicine | 3.3M | 30 | U | P | Y |
| PhilPapers Taxonomy | Philosophy | 7447 | 5 | T | P | N |
| Physics Subject Headings | Physics | 3749 | 6 | O | P | Y |
| Physics and Astronomy Classif. Scheme | Physics | 8260 | 5 | T | M | N |
| PsycInfo and PsycTests Classification Systems | Psychology | 188 | 3 | T | M | N |
| TheSoz | Social Science | 8223 | 6 | O | P | Y |
| All Science Journal Classification Codes | Multi-field | 334 | 2 | T | M | N |
| ArXiv Subjects | Multi-field | 176 | 3 | T | M | N |
| Dewey Decimal Classification | Multi-field | 60K | 14 | T | P | Y |
| DFG Classification | Multi-field | 278 | 4 | T | M | N |
| European Commission Taxonomy | Multi-field | 629 | 3 | T | M | N |
| European Research Council Taxonomy | Multi-field | 431 | 3 | T | M | N |
| EuroVoc | Multi-field | 7439 | 6 | O | P | Y |
| Fields of Research (ANZSRC) | Multi-field | 4406 | 3 | T | M | N |
| Fields of research and development (OECD) | Multi-field | 48 | 2 | T | M | N |
| KNOWMAK | Multi-field | 72 | 3 | O | P | Y |
| Library of Congress Class. (and Subj. Head.) | Multi-field | 467K | 29 | O | P | Y |
| Microsoft's Fields of Study | Multi-field | 704K | 6 | T | P | N |
| Modern Science Ontology | Multi-field | 369 | 6 | O | P | N |
| Nature Subjects | Multi-field | 2852 | 8 | O | P | N |
| OpenAIRE's Field of Science Taxonomy | Multi-field | 50K | 6 | T | P | N |
| OpenAlex Topics | Multi-field | 4798 | 4 | T | M | N |
| Research Fields, Courses and Disciplines (ASRC) | Multi-field | 1061 | 3 | T | M | N |
| Science Metrix Classification | Multi-field | 199 | 3 | T | M | N |
| Socio-Economic Objective (ANZSRC) | Multi-field | 1974 | 3 | T | M | N |
| Subject Resource Application Ontology | Multi-field | 435 | 8 | O | P | Y |
| UNESCO Thesaurus | Multi-field | 4482 | 6 | O | P | Y |
| Web of Science Categories | Multi-field | 254 | 1 | L | - | N |

*Salatino et al. (2025) "**A survey of knowledge organization systems of research fields: Resources and challenges**." Quantitative Science Studies 6: 567-610.*




# Topic Classification for UKRI

Rick Payne and team / Better Images of AI / AI is... / CC-BY 4.0

# Tracking Stars and Unicorns

- ~£310K project funded by the UK Metascience Unit, through the Metascience Sandpit 2025
- Understanding how research funding shapes both people and ideas over time.
- Funders need evidence on how funding pathways support sustainable research careers (**stars**) while enabling novel, high-potential ideas (**unicorns**) to emerge and flourish.

- ***This presentation focuses on Unicorns.***

# Data

- UKRI Funding Application Data
  - **85**K funded applications
  - **258**K unfunded applications
  - Within a Trusted Research Environments (TRE)

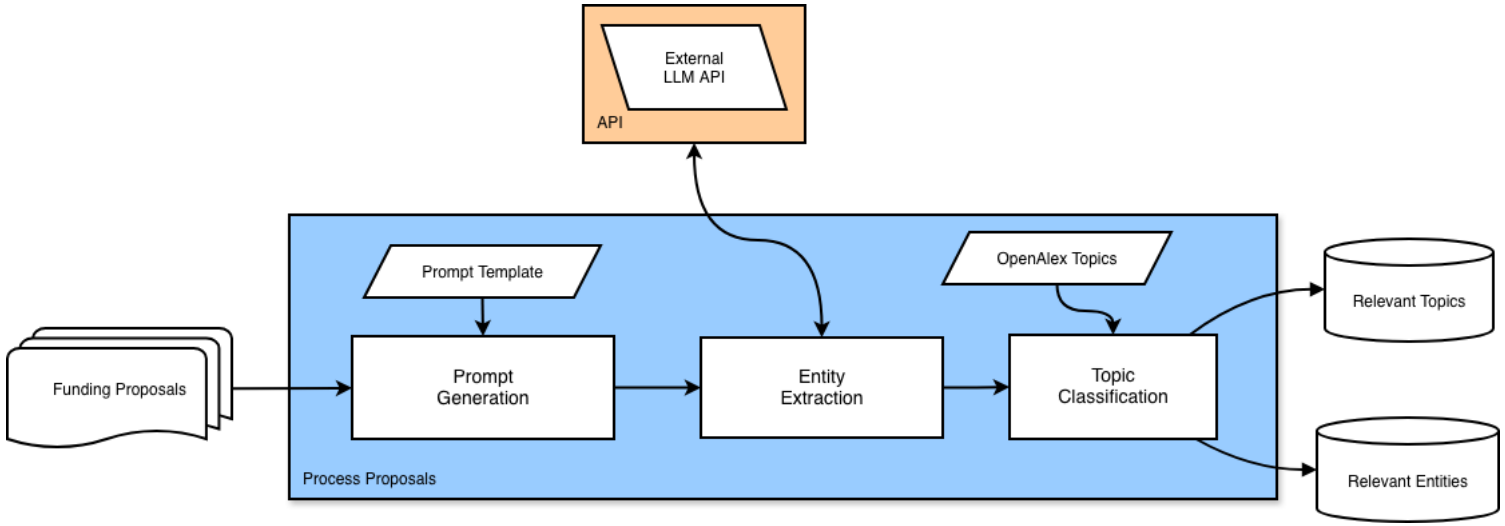
- Gateway to Research
  - Platform for accessing funded proposals and related research outputs
  - Scraped from https://gtr.ukri.org/ *

- Data spans the seven statutory research councils and a specialist national centre forming UKRI: AHRC, BBSRC, MRC, ESRC, NERC, STFC, EPSRC, NC3Rs

* Filgueira, Yu, Ruan, et al. ImpactLens: Using AI and Data to Communicate Research Impact Effectively. In 2025 IEEE International Conference on eScience (eScience)

# Approach

1. Extracting relevant keywords/research entities
2. Topic Classification
    - OpenAlex Topics




Ruan, X., Salatino, A., Filgueira, R., Moraw, K., Marcoci, A., Derrick, G., & Callaghan, S. (2026). **Research Entity Extraction and Topic Detection from UKRI Grant Proposals**. *arXiv preprint arXiv:2606.30304*.

# Extracting Research Entities

> A ***research entity*** is defined as a discrete semantic unit representing a functional component of a scientific inquiry, including thematic topics, disciplines, methodologies, and technical facets such as software and datasets.

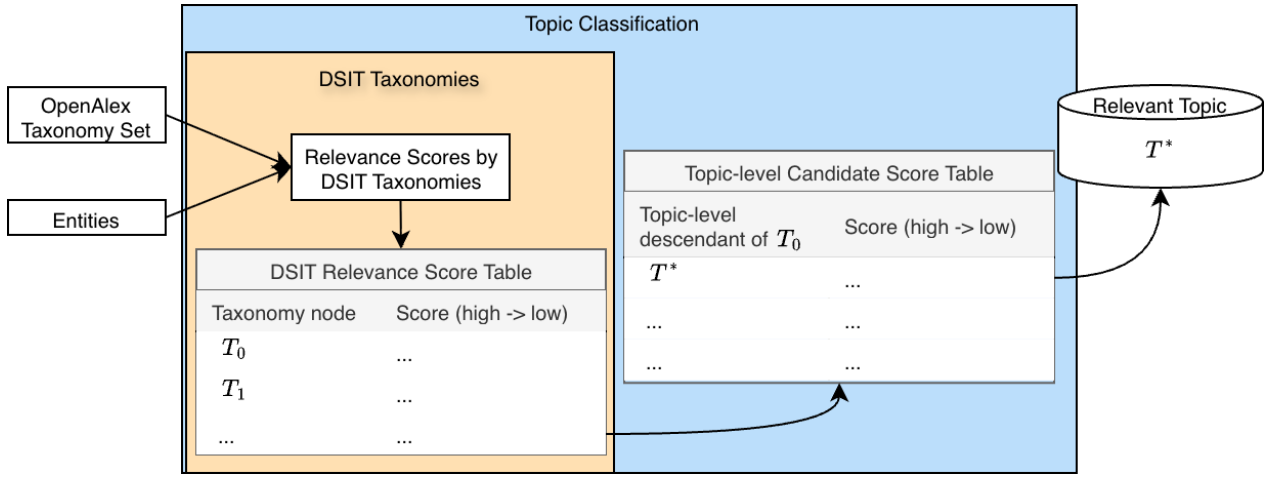
Employed a prompt template to extract:

- **research topics** (E.g., "social impact of technology", "urban heat island effect”)
- **discipline** (E.g., "Computational Linguistics", "developmental psychology”)
- **methodologies** (E.g., "Gas Chromatography", "content analysis", "statistical modelling")
- **approaches** (E.g., "case study", "experimental design", "systematic review”)
- **tools** (E.g., "mass spectrometer", "telescope arrays", "Scanning Electron Microscope")
- **dataset** (E.g., "demographic census data", "patient electronic health records”)
- **software** (E.g., "R statistical package", "MATLAB", "secure blockchain platform”)
- **artefacts** (E.g., "simulation model", "policy recommendation report", "code repository")
- **events/occurrences** (E.g., "2008 financial crisis", "COVID-19 pandemic", "September 11")

Used Mistral-7B-Instruct-v0.3

# Topic Classification

- Map each proposal to an OpenAlex topic using the extracted entities
- **DSIT-Taxonomies (++)**
- ***ORIGINAL***: matches entities, sentences, and whole abstract to a relevant node in the taxonomy
  - Matches done can appear at any level: domain, field, subfield, topic.
- ***EXTENDED***: ensures it matches a topic, and we extracts the taxonomical branch




https://github.com/innovation-growth-lab/dsit-taxonomies

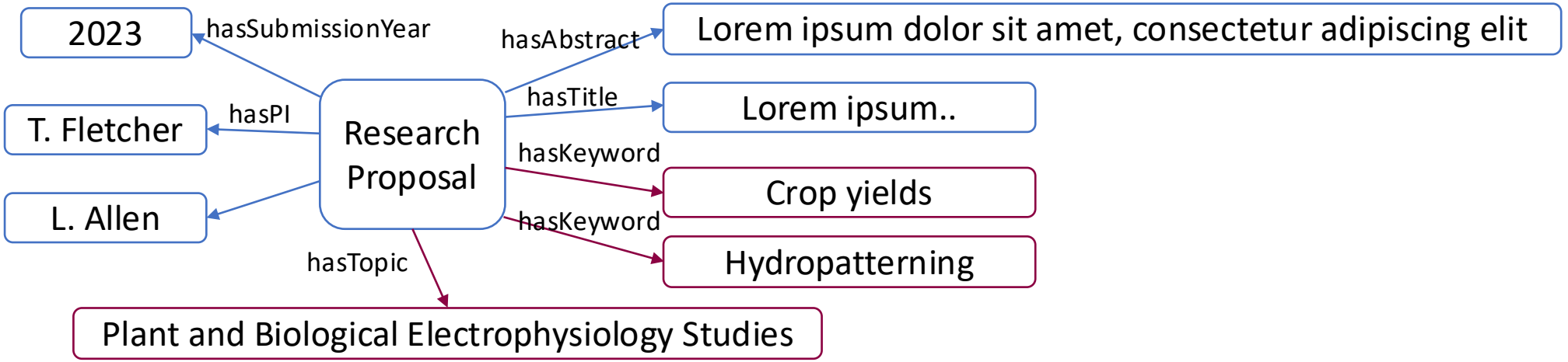
# Results

- Manual evaluation
  - Accuracy of the method is ~90%
  - Depending on the task (how the data will be aggregated) this is a good result.
  - Compared against GPT4 and DSIT-Taxonomies

- LLMs are good at identifying research entities:
  - They can still hallucinate, but easily removed in post-processing
  - Whereas traditional keyword extractors:
    - Include **verbs** (e.g., “provide”), **quantifiers** (e.g., “thousands”), and **adverbs** (e.g., “however”)
    - Breaks concepts: “signal” and “amplifier” instead of “signal amplifier”
    - ***Noisy output***

# What happened in the meantime?

- We applied this algorithm to the whole UKRI dataset
  - ~340K applications within the Trusted Research Environment




- This is allowing us to run longitudinal analyses
- Fine-grained trend analyses
- Benchmarking domestic progress against global competitors (via OpenAlex)

# Lesson learnt

- Knowledge Organization Systems need to be chosen based on:
  - Use cases (cross-domain, depth, is it for archival purposes?)
  - Interoperability
  - Requirements
- Quality of topic extraction needs to reflect the use case too
- LLMs are good but quite expensive if run at large scale and **slow**!
  - Perhaps we can take advantage of LLMs to write better and refined strategies based on the technologies we have
- In topic classification, topics are retrofitted
  - This may rule out some analyses

# References


- *Salatino et al. (2026) "Are Researchers Being Replaced by Artificial Intelligence?." arXiv preprint arXiv:2605.16294. Ruan,*
- *Xingran, et al. "Research Entity Extraction and Topic Detection from UKRI Grant Proposals." arXiv preprint arXiv:2606.30304 (2026).*
- *Salatino et al. (2026) "Does Diversity of Expertise Drive Citation Impact? Evidence from Computer Science." Scientometrics: 1-28.*
- *Salatino et al. (2025) "A survey of knowledge organization systems of research fields: Resources and challenges." Quantitative Science Studies 6: 567-610.*
- *Meloni et al. (2023) "Integrating conversational agents and knowledge graphs within the scholarly domain." Ieee Access 11: 22468-22489.*
- *Angioni et al. (2022) "Leveraging Knowledge Graph Technologies to Assess Journals and Conferences at Springer Nature." ISWC 2022.*
- *Angioni et al. (2021). "AIDA: A knowledge graph about research dynamics in academia and industry." Quantitative Science Studies, 2, 4.*
- *Salatino et al. (2020) "Researchflow: Understanding the knowledge flow between academia and industry."* EKAW 2020.
- *Salatino et al.* (2019) *"The CSO classifier: Ontology-driven detection of research topics in scholarly articles."* TPDL 2019
- *Salatino et al. (2018) "The computer science ontology: a large-scale taxonomy of research areas."* ISWC 2018


SKM3

KMi Knowledge Media Institute

The Open University


angelo.salatino@open.ac.uk

@angelosalatino.bsky.social